Task-2 SubTask-1

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I. Introduction

Road extraction from aerial imagery is a fundamental task in remote sensing and computer vision. Accurate road segmentation enables applications such as navigation systems, map generation, infrastructure monitoring, and autonomous transportation.

Recent advances in deep learning have significantly improved semantic segmentation performance. Encoder-decoder architectures such as U-Net have become popular due to their ability to preserve spatial information while extracting high-level features. More advanced architectures including DeepLabV3+ and transformer-based models have further improved segmentation capabilities in many benchmark datasets.

This project investigates the effectiveness of several segmentation architectures for road extraction from aerial images and compares their performance under identical preprocessing and training conditions.

II. Dataset

The Massachusetts Roads Dataset was used for all experiments. The dataset contains aerial images along with corresponding binary road masks.

Dataset structure:

- Train Images
- Validation Images
- Test Images
- Corresponding Binary Road Masks

Road pixels occupy only a small fraction of the image area, creating a significant class imbalance problem.

III. Data Preprocessing

A preprocessing pipeline was developed to standardize the dataset before training.

The following steps were performed:

1) Resize all images to 512×512 pixels.
2) Resize masks using nearest-neighbor interpolation.
3) Convert TIFF images to PNG format.
4) Store processed data in dedicated train, validation, and test folders.

Image interpolation used Lanczos resampling while masks used nearest-neighbor interpolation to preserve binary labels.

IV. Loss Function

Road segmentation suffers from severe class imbalance because most pixels belong to the background class.

To address this issue, a combination of Binary Cross Entropy (BCE) Loss and Dice Loss was used:

$$L = L_{BCE} + L_{Dice} \tag{1}$$

Binary Cross Entropy provides stable pixel-wise supervision while Dice Loss directly optimizes overlap between predicted masks and ground-truth masks.

Dice coefficient is defined as:

$$Dice = \frac{2|P \cap G|}{|P| + |G|} \tag{2}$$

where P represents predicted pixels and G represents ground-truth pixels.

V. Evaluation Metrics

Two primary metrics were used:

A. *Dice Score*

Measures overlap between predicted and ground-truth masks.

B. *Intersection over Union (IoU)*

$$IoU = \frac{|P \cap G|}{|P \cup G|} \tag{3}$$

IoU provides a stricter evaluation by measuring the ratio of intersection to union.

VI. Baseline U-Net

The baseline model uses a standard U-Net encoder-decoder architecture with skip connections.

Encoder feature dimensions:

$$64 \rightarrow 128 \rightarrow 256 \rightarrow 512$$

Skip connections preserve spatial information and help reconstruct fine road structures during decoding.

Training configuration:

- Input Size: 512×512
- Batch Size: 8
- Learning Rate: 1×10^{-4}
- Epochs: 10
- Optimizer: Adam
- Loss: BCE + Dice

VII. Architectural Improvements

Three additional architectures were evaluated:

A. *ResNet34 U-Net*

The encoder was replaced with a pretrained ResNet34 backbone initialized using ImageNet weights.

B. *DeepLabV3+*

DeepLabV3+ introduces Atrous Spatial Pyramid Pooling (ASPP) to capture multi-scale contextual information.

C. SegFormer

SegFormer is a transformer-based semantic segmentation model that uses hierarchical transformer encoders for feature extraction.

VIII. Results

TABLE I
MODEL COMPARISON

Model	Dice Score	IoU
U-Net	0.7245	0.5690
ResNet34 U-Net	0.7180	0.5612
DeepLabV3+	0.6093	0.4383
SegFormer	0.5726	0.4012

IX. Training Curves

Figure 1 shows the training and validation loss curves.

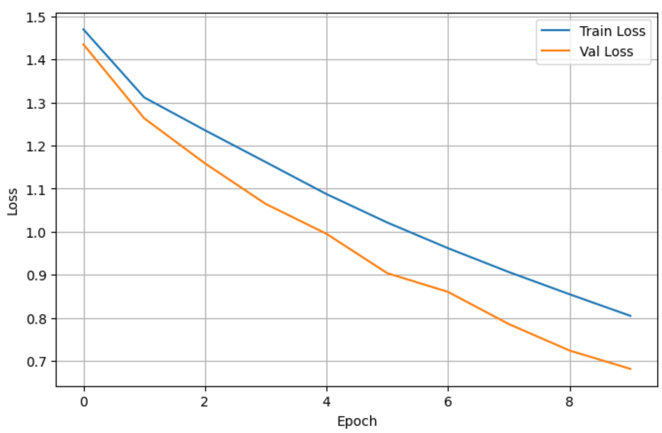

Fig. 1. Training and Validation Loss

Figure 2 shows Dice and IoU progression during training.

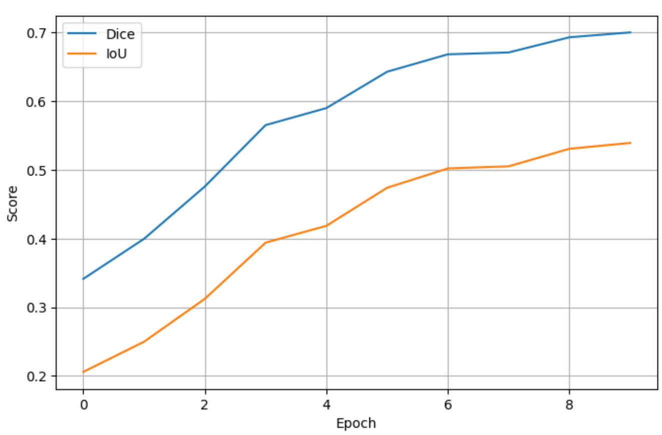

Fig. 2. Dice and IoU Progression

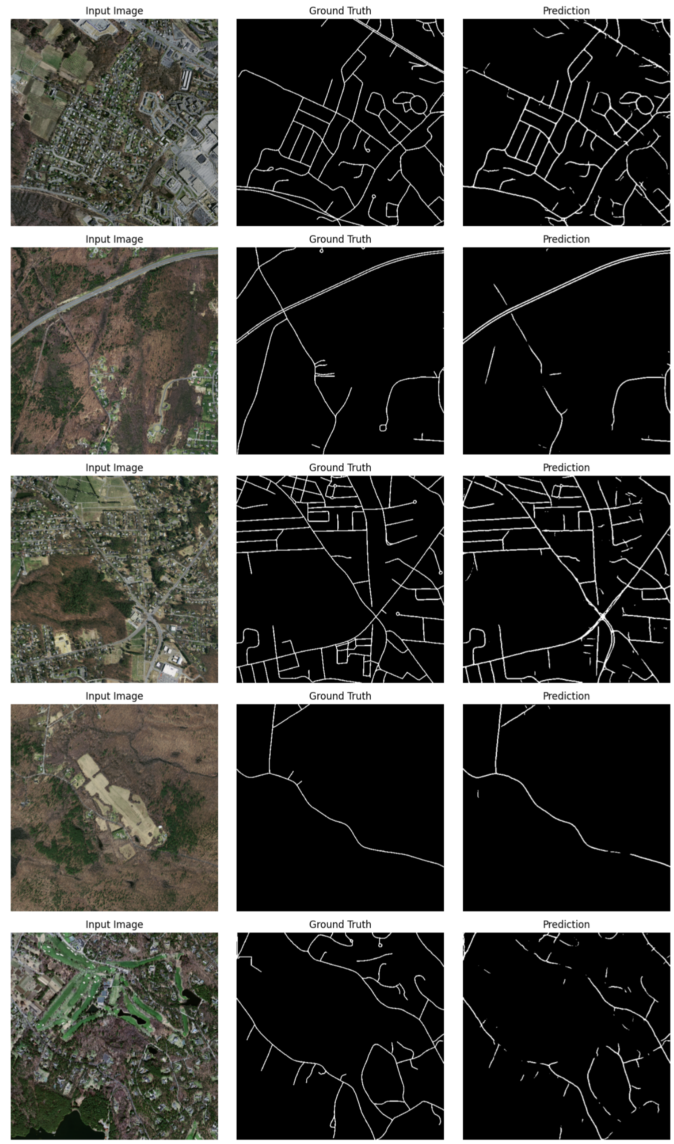

Fig. 3. Sample Predictions

X. Qualitative Results

Example predictions from the baseline U-Net are shown in Figure 3.

The model successfully captures major road structures and intersections. Most errors occur on thin residential roads and small road branches.

XI. Discussion

The baseline U-Net achieved the best performance despite being the simplest architecture evaluated.

ResNet34 U-Net converged faster during training but did not improve final accuracy. DeepLabV3+ and SegFormer underperformed under the chosen training budget. This suggests that the Massachusetts Roads Dataset favors architectures that preserve fine spatial details through skip connections.

The combination of BCE and Dice Loss proved effective in handling class imbalance and improving segmentation quality.

XII. Conclusion

This project evaluated multiple deep learning architectures for road segmentation using the Massachusetts Roads Dataset.

The baseline U-Net achieved the best performance with:

- Dice Score: 0.7245
- IoU: 0.5690

Experimental results indicate that a carefully trained U-Net remains highly effective for aerial road segmentation and can outperform more sophisticated architectures under identical training conditions.